HK IMO Training 2023-2024
Problem Set 5 Suggested Solutions

Problem 17 (Estonia TST 2019 Problem 3)**.** Find all function(s) $f:\mathbb{R}\to\mathbb{R}$ such that

$$f(x^2)f(y^2)+|x|\,f(-xy^2)=3\,|y|\,f(x^2y) \tag{1}$$

for any $x,y\in\mathbb{R}$.

Solution. Answer: $f(x)=0$ or $f(x)=2\,|x|$.

Replacing x by $-x$ in (1), we get

$$f(x^2)f(y^2)+|x|\,f(xy^2)=3\,|y|\,f(x^2y).$$

Comparing this with (1), we see that $|x|\,f(-xy^2)=|x|\,f(xy^2)$. For any $x\neq 0$, this yields $f(-x)=f(x)$ by putting $y=1$. Therefore, f is an even function. Thus, it suffices to consider $x,y\geqslant 0$. In that case, (1) becomes

$$f(x^2)f(y^2)+xf(xy^2)=3yf(x^2y). \tag{2}$$

Swapping x and y, we obtain

$$f(x^2)f(y^2)+yf(x^2y)=3xf(xy^2). \tag{3}$$

Taking the difference of (2) and (3), we deduce $4xf(xy^2)=4yf(x^2y)$, i.e.

$$xf(xy^2)=yf(x^2y). \tag{4}$$

Define u and v such that $u=xy^2$ and $v=x^2y$ for nonzero x,y. Equivalently, $x=u^{-\frac{1}{3}}v^{\frac{2}{3}}$ and $y=u^{\frac{2}{3}}v^{-\frac{1}{3}}$. Then (4) becomes $u^{-\frac{1}{3}}v^{\frac{2}{3}}f(u)=u^{\frac{2}{3}}v^{-\frac{1}{3}}f(v)$, i.e.

$$\frac{f(u)}{u}=\frac{f(v)}{v}.$$

Therefore, there exists a constant c such that $f(u)=cu$ for all $u>0$. Putting $x=0$ and $y=1$ in (4), we have $f(0)=0$. Thus, $f(u)=cu$ holds for all $u\geqslant 0$. Putting this into (2), we need

$$c^2x^2y^2+cx^2y^2=3cx^2y^2$$

for all $x,y\geqslant 0$. This holds if and only if $c^2+c=3c$, i.e. $c=0,2$. This means $f(x)=0$ for all $x\geqslant 0$ or $f(x)=2x$ for all $x\geqslant 0$. As f is even, $f(x)=0$ for all $x\in\mathbb{R}$ or $f(x)=2\,|x|$ for all $x\in\mathbb{R}$.

Finally, we check that both are solutions. For $f(x)=0$, both sides of (1) are obviously 0. For $f(x)=2\,|x|$, we have

$$f(x^2)f(y^2)+|x|\,f(-xy^2)=4x^2y^2+2x^2y^2=3(2x^2y^2)=3\,|y|\,f(x^2y).$$

□

Problem 18 (Denmark MO Mohr Contest 2023 Problem 5)**.** There is a circular chessboard with 100 cells labelled $0,1,2,\ldots,99$ arranged in this order. We put a chess piece in the cell with label 0. We choose three distinct positive integers a,b,c from $1,2,\ldots,99$ and carry out the following operations in order:

- Move the chess piece a cells forward and put a coin in the cell that it lands on. Repeat this 33 times.
- Move the chess piece b cells forward and put a coin in the cell that it lands on. Repeat this 33 times.
- Move the chess piece c cells forward and put a coin in the cell that it lands on. Repeat this 33 times.

At most how many coins can we put in the cell with label 1 at the end?

Solution. Answer: 25.

For example, consider $a=97$, $b=50$ and $c=25$. Then the cells that the chess piece lands on are

$$0,97,94,91,88,\ldots,1,$$
$$51,1,51,1,51,\ldots,51,$$
$$76,1,26,51,76,\ldots,76$$

in this order. In the first 33 steps (stage I), it only visits cell 1 once. In the next 33 steps (stage II), it visits cell 1 for $\left\lfloor \frac{33}{2} \right\rfloor = 16$ steps. In the last 33 steps (stage III), it visits cell 1 for $\left\lfloor \frac{33}{4} \right\rfloor = 8$ steps. There are $1 + 16 + 8 = 25$ coins in cell 1 at the end.

We now show that there are at most 25 coins. Consider a particular stage. Initially, the chess piece stays in cell k. Then it visits the cells $k + d, k + 2d, \ldots, k + 33d$ in this stage where all labels are taken modulo 100. The labels are periodic, where the period is $s = \frac{100}{(d, 100)}$. Also, there can be at most $\left\lceil \frac{33}{s} \right\rceil$ cells with label 1.

- As $1 \leqslant d \leqslant 99$, the smallest possible value of s is 2. This only holds when $d = 50$. There can be at most 17 cells with label 1 in this stage. The maximal case is attained only when $k = 1$ (and $k + 33d \equiv 1 \pmod{100}$).
- As $s \mid 100$, the next smallest possible value of s is 4. This only holds when $d = 25, 75$. There can be at most 9 cells with label 1 in this stage. The maximal case is attained only when $k = 1$ (and $k + 33d \equiv 1 \pmod{100}$).
- For all other values, we have $s \geqslant 5$. There can be at most 7 cells with label 1.

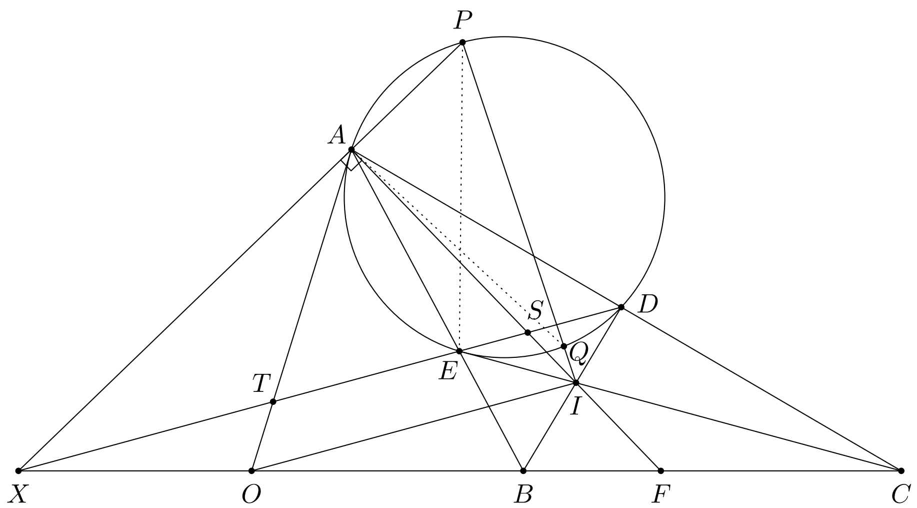
However, for stage I, we have $k = 0$. Since $jd \equiv 1 \pmod{100}$ iff $(d, 100) = 1$ and $j \equiv d^{-1} \pmod{100}$, there can be at most one cell with label 1. Also, if there are 17 and 9 (or 9 and 17) cells with label 1 in stages II and III respectively, then the last cell visited in stage II and the first cell visited in stage III are both 1, which is impossible. Therefore, there can be at most $1 + (17 + 9 - 1) = 26$ cells with label 1.

If there are exactly 26 cells with label 1, then $(b, c) = (50, 25), (50, 75), (25, 50), (75, 50)$. Also, the last cell visited in the stage I can only be cell $1, 26, 51, 76$.

- If this cell has label 1, then the first cell visited in stage II is $1 + b \not\equiv 1 \pmod{100}$, and the first cell visited in stage III is $1 + 33b + c \equiv 1 + b + c \not\equiv 1 \pmod{100}$. This contradicts to the assumption that there are $17 + 9 - 1$ cells with label 1 in stages II and III.
- If this cell has label $26, 51, 76$, then $33a \equiv 26, 51, 76 \pmod{100}$, and hence $a = 22, 47, 72 \pmod{100}$ respectively. Since $22j \equiv 1 \pmod{100}$ and $72j \equiv 1 \pmod{100}$ have no solution, while $47j \equiv 1 \pmod{100}$ holds iff $j \equiv 83 \pmod{100}$, there is no $j = 1, 2, \ldots, 33$ such that $ja \equiv 1 \pmod{100}$. This is a contradiction.

Therefore, there are at most 25 coins in cell 1. $\square$

Problem 19 (Israel TST 2022 Test 7 Problem 3). Let I be the incentre of $\triangle ABC$. The angle bisectors of $\angle B$ and $\angle C$ meet the opposite sides at D and E respectively. Let P be the point on the circumcircle ω of $\triangle ADE$ such that $\angle PAI = 90°$. The line PI intersects ω again at Q. Prove that the circumcentre of $\triangle AIQ$ lies on the line BC.



Solution. Let AI meet BC at F. Let O be the point on BC such that $OI /\!/ ED$. Let DE meet BC, AF, AO at X, S, T respectively. Since AF, BD, CE are concurrent, we have $(X, F; B, C) = -1$. Using the projection

$$(X, F, B, C) \xrightarrow{D} (S, F, I, A) \xrightarrow{O} (S, X, \infty_{DE}, T),$$

we see that $(S, X; \infty_{DE}, T) = -1$, and hence T is the midpoint of SX.

Using another projection

$$(X, F, B, C) \xrightarrow{A} (X, S, E, D),$$

we see that $(X, S; E, D) = -1$. As AS bisects $\angle EAD$, we have $\angle XAS = 90^\circ$. Therefore, $TX = TS = TA$, and hence $\angle TAS = \angle TSA = \angle OIA$. Thus, $OI = OA$.

Next, note that P is the midpoint of the arc DE on (ADE) as AI bisects $\angle DAE$ and $\angle PAI = 90^\circ$. Then we find that

$$\angle PEA = \angle DEA - \angle DEP = \angle DEA - \frac{180^\circ - \angle EPD}{2} = \angle DEA - 90^\circ + \frac{A}{2} = \angle ESI - 90^\circ = 90^\circ - \angle AIO = \frac{1}{2}\angle IOA.$$

Using directed angles, we have

$$2\measuredangle IQA = 2\measuredangle PQA = 2\measuredangle PEA = \measuredangle IOA.$$

Together with $OI = OA$, we see that O is the circumcentre of $\triangle AIQ$. This completes the proof. $\square$

Problem 20 (Balkan MO Shortlist 2018 N5 modified)**.** Find all positive integers m and n such that

$$k^2m^2 + 1 \mid n^{\varphi(k)} - 1$$

for every positive integer k.

Solution. Answer: $n = 1$ and m can be any positive integer.

Clearly, $n = 1$ is a solution as $n^{\varphi(k)} - 1 = 0$ is divisible by anything. It remains to prove $n = 1$ is the only possibility. Consider $k = 2^t$. Then we have

$$2^{2t}m^2 + 1 \mid n^{2^{t-1}} - 1$$

for all t. For any fixed m, let s be a sufficiently large positive integer.

Claim. There are infinitely many primes p dividing $2^{2t}m^2 + 1$ for some t, such that $2^{2s} \nmid p - 1$.

Suppose on the contrary that only the primes $p_1, p_2, \ldots, p_\ell$ satisfy these properties. Clearly, each p_i is odd. We can let

$$N := 2^2m^2 + 1 = p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_\ell^{\alpha_\ell}q,$$
$$2^{2(1+sp_1p_2\cdots p_\ell\varphi(N))}m^2 + 1 = p_1^{\beta_1}p_2^{\beta_2}\cdots p_\ell^{\beta_\ell}r,$$

where all prime divisors of q and r are congruent to 1 modulo 2^{2s}. As $\varphi(p_i^{\alpha_i+1}) \mid p_i\varphi(N)$, we have

$$2^{2(1+sp_1p_2\cdots p_\ell\varphi(N))}m^2 + 1 \equiv 2^2m^2 + 1 \pmod{p_i^{\alpha_i+1}}$$

for every i. This implies $\beta_i = \alpha_i$. Thus,

$$N \equiv p_1^{\alpha_1}p_2^{\alpha_2}\cdots p_\ell^{\alpha_\ell} \equiv 2^{2(1+sp_1p_2\cdots p_\ell\varphi(N))}m^2 + 1 \equiv 1 \pmod{2^{2s}}.$$

But this is impossible as $N = 4m^2 + 1$ is fixed, while s is sufficiently large. This proves our claim.

By the claim, there exists a sufficiently large prime p which divides $2^{2t}m^2 + 1$ for some t and $2^{2s} \nmid p - 1$ (where s is fixed). For this t, we have $p \mid n^{2^{t-1}} - 1$, so that $n^{2^{t-1}} \equiv 1 \pmod{p}$. Let d be the order of n modulo p. We must have $d \mid 2^{t-1}$. Thus, $d = 2^h$ for some $h \leqslant t - 1$. Also, as $d \mid p - 1$ and $2^{2s} \nmid p - 1$, we must have $h \leqslant 2s - 1$. Therefore, $n^{2^{2s-1}} \equiv 1 \pmod{p}$. Since p is sufficiently large, the only possibility is $n = 1$ as desired. $\square$